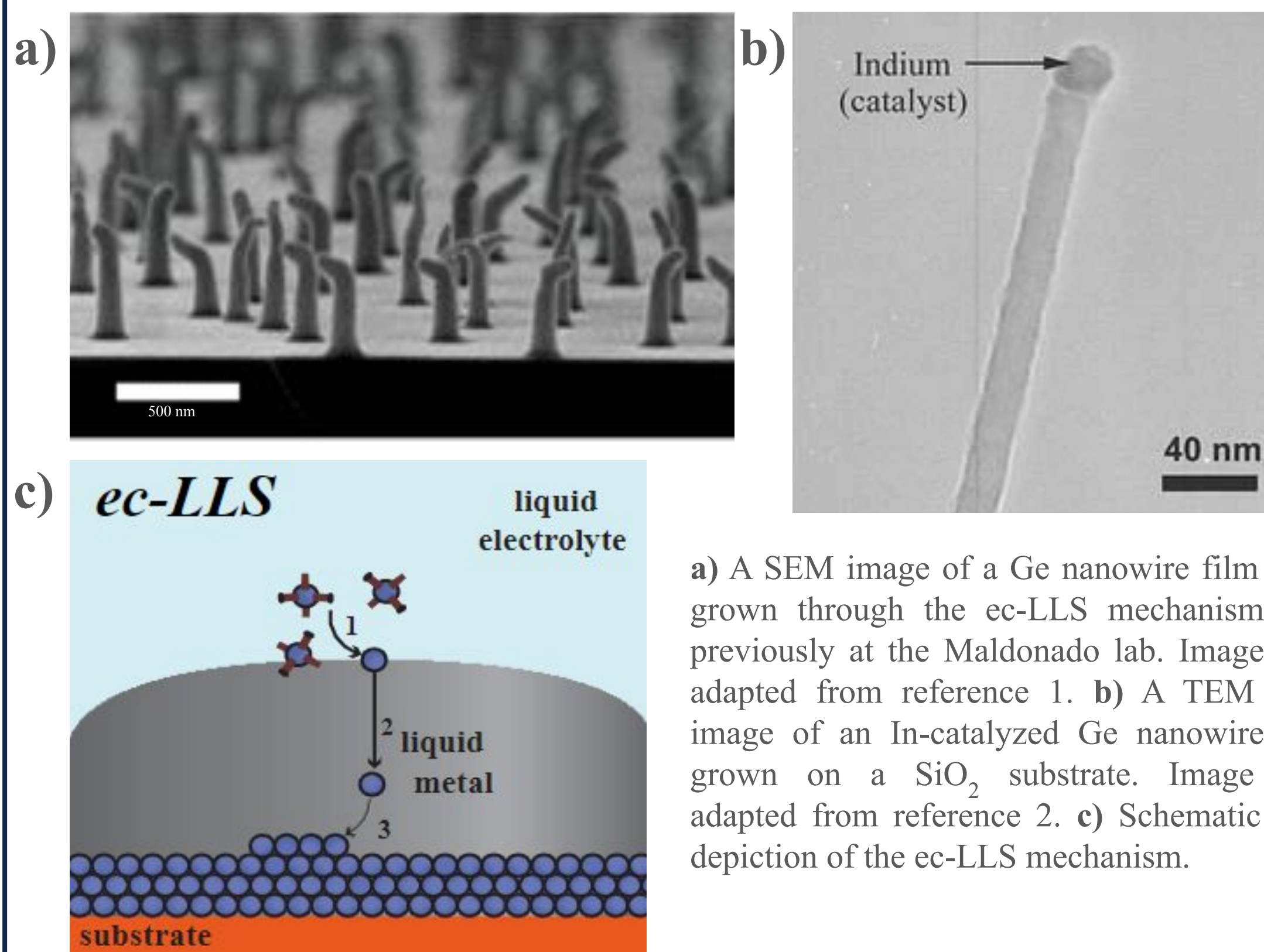




Introduction

Why are Ge nanowires important?

- Crystalline covalent semiconductors are widely used in battery anodes, solar cells, and circuits. Nanowires are used in nanoelectronics and optoelectronic devices.
- The electrochemical liquid-liquid-solid (ec-LLS) crystal growth mechanism requires less energy and resources than traditional means of synthesizing semiconductor crystals.
- ec-LLS Mechanism:** 1) Ge^{4+} salt in solution is reduced. 2) Reduced Ge dissolves into the eutectic droplet. 3) Once supersaturated, Ge crystals form on substrate.



References

- Fahrenkrug, E.; Maldonado, S. Electrochemical Liquid-Liquid-Solid (Ec-LLS) Crystal Growth: A Low-Temperature Strategy for Covalent Semiconductor Crystal Growth. *Acc. Chem. Res.* **2015**, *48* (7), 1881–1890. <https://doi.org/10.1021/acs.accounts.5b00158>.
- Yu, B.; Sun, X. H.; Calebotta, G. A.; Dholakia, G. R.; Meyyappan, M. One-Dimensional Germanium Nanowires for Future Electronics. *J. Clust. Sci.* **2006**, *17* (4), 579–597. <https://doi.org/10.1007/s10876-006-0081-x>.

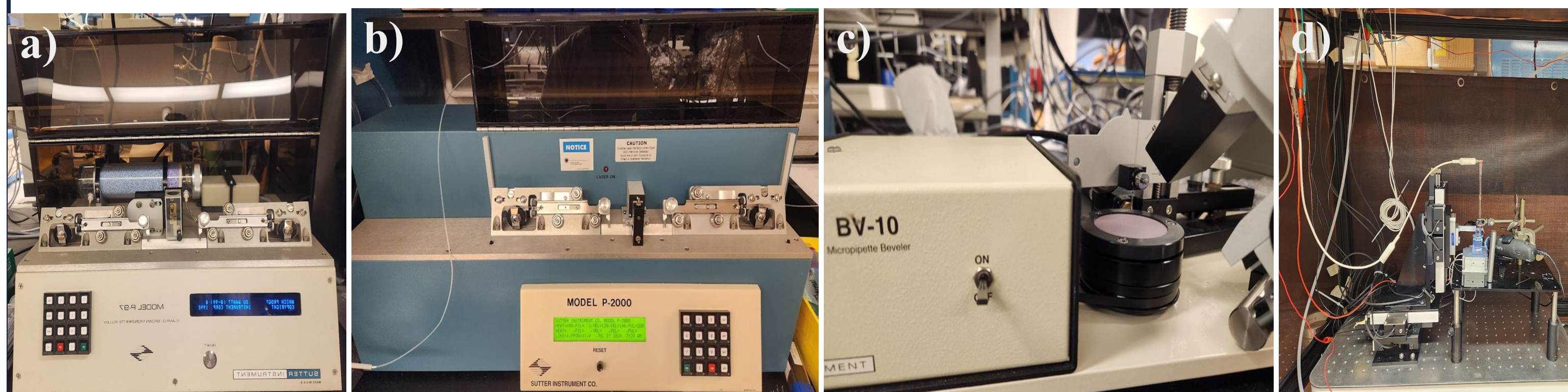
Acknowledgments

- Lauren C. Rich
- Dr. Stephen Maldonado and the Maldonado lab
- University of Michigan Chemistry REU Program



Methods

Instruments



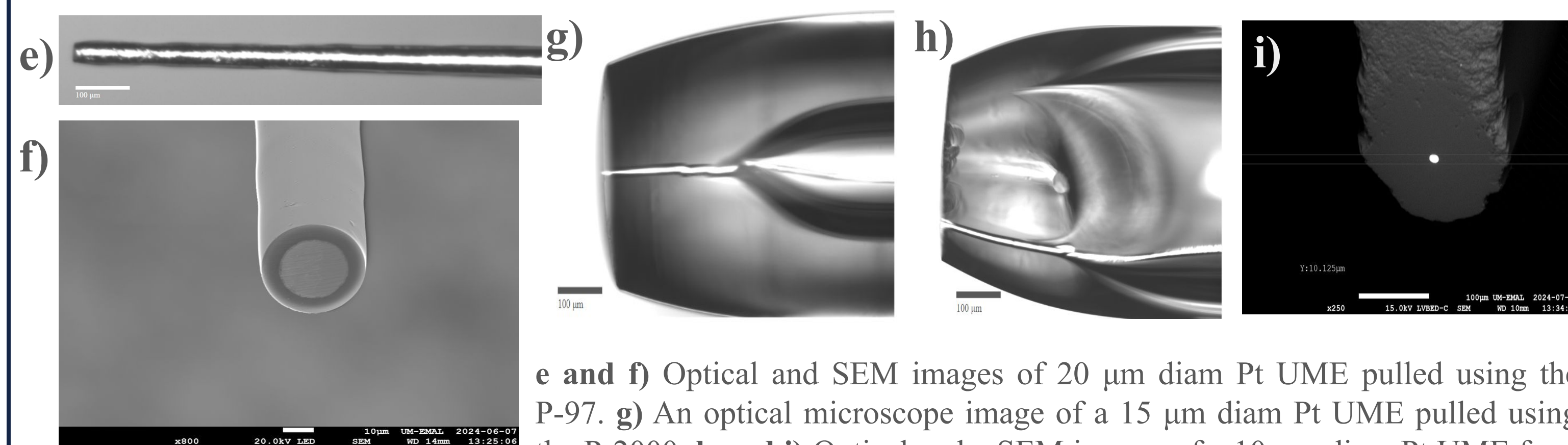
Images of the instruments used in the Maldonado lab: **a)** The P-97 Flaming/Brown Micropipette Puller. **b)** The P-2000 Laser-Based Micropipette Puller. **c)** The BV-10 Microelectrode Beveler. **d)** The SEPM instrument.

Pulling UMEs

General Procedure: 25 μm diameter wires (either Pt or Cu) were inserted into glass capillaries before being secured in either the P-97 or the P-2000. The capillaries were first sealed and then hard pulled, resulting in two ultramicroelectrodes (UMEs).

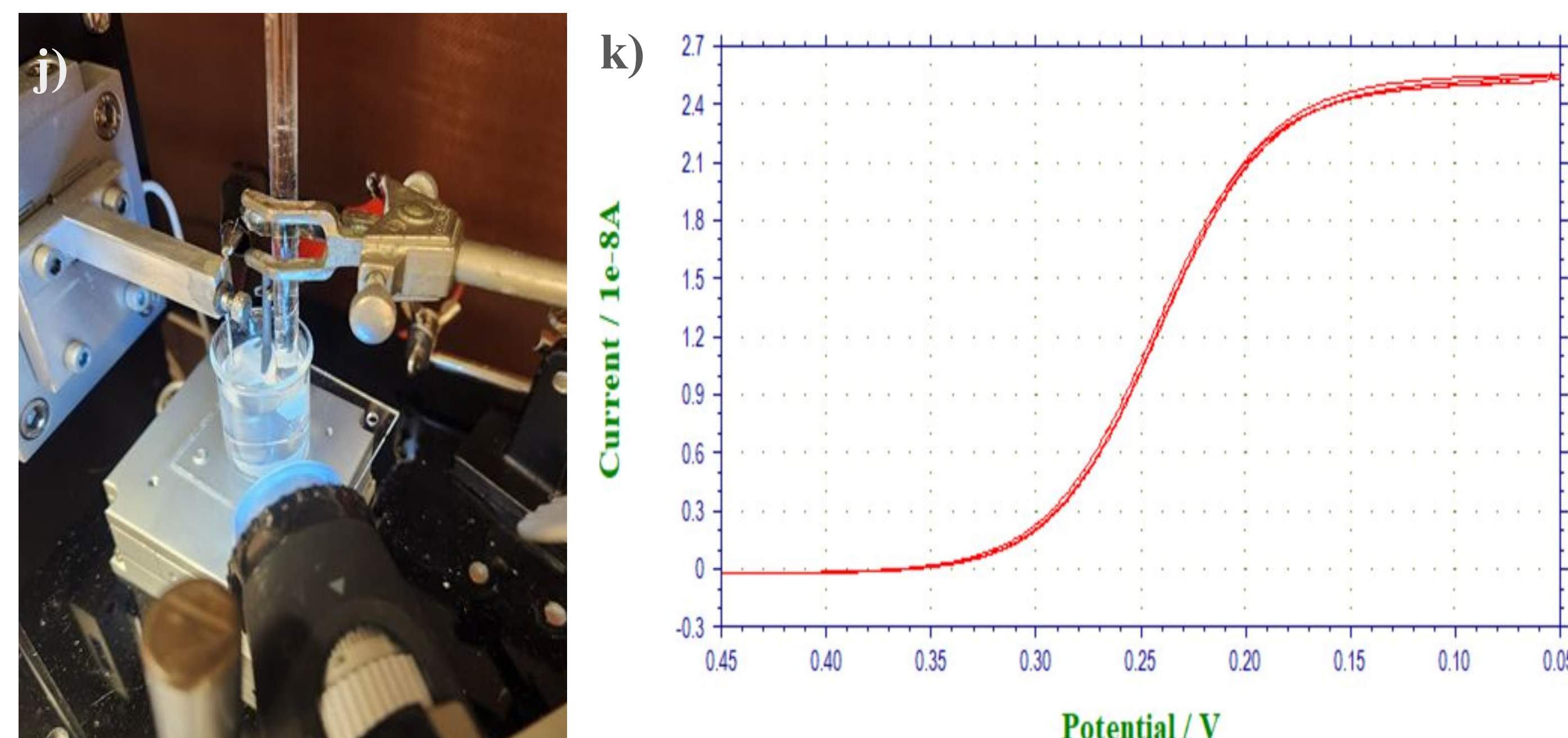
P-97: single or separate programs

P-2000: separate sealing and pulling programs

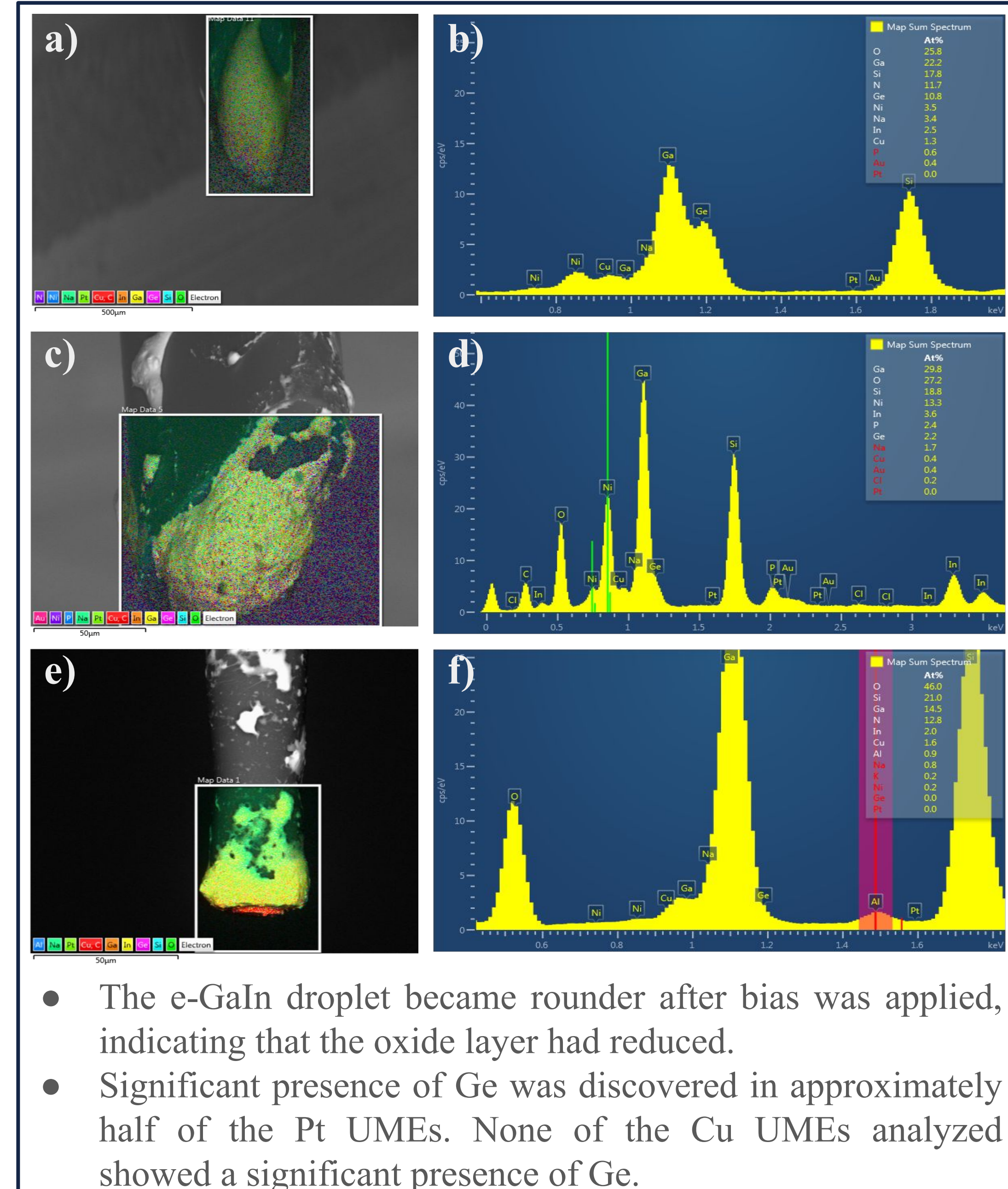


Characterizing UMEs

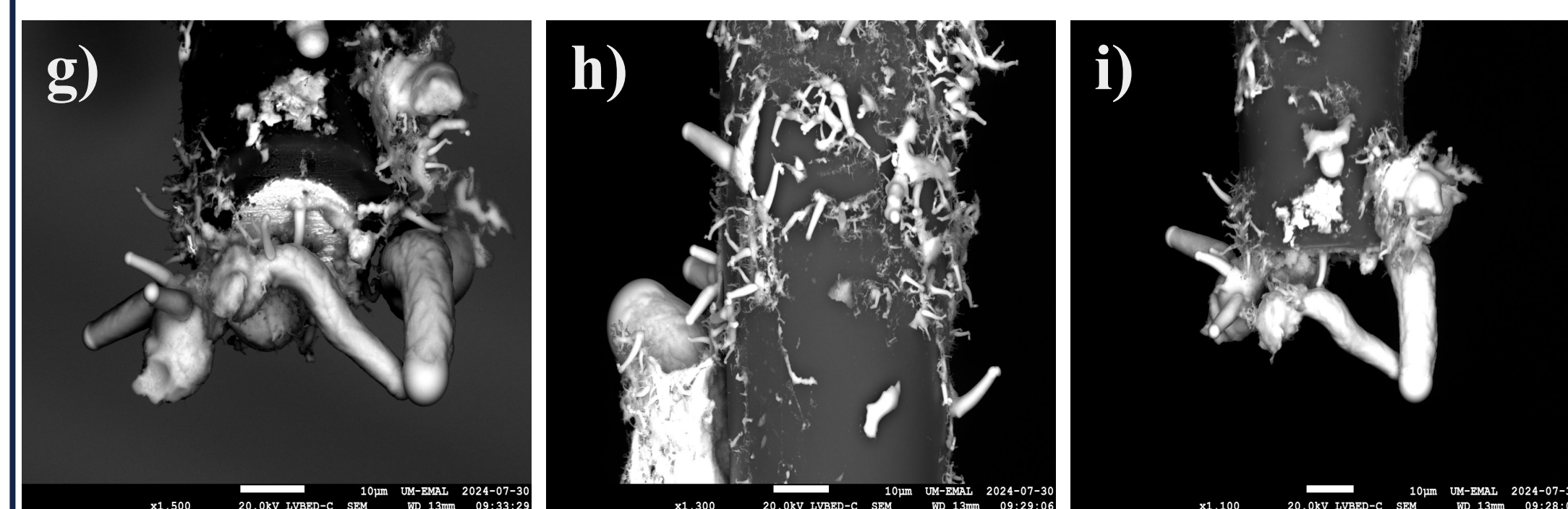
Data from cyclic voltammograms was collected and sigmoidally fitted using Origin Pro to determine the steady-state current (i_{ss}) using the formula $i_{ss} = 4\pi nFD_0C_0^*r_0$. An average radius of 16.7 μm was calculated.



Results



- The e-GaIn droplet became rounder after bias was applied, indicating that the oxide layer had reduced.
- Significant presence of Ge was discovered in approximately half of the Pt UMEs. None of the Cu UMEs analyzed showed a significant presence of Ge.



Where is the Ge being Deposited?

